Introduction to Software Engineering

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1. System Vision Document

Vision Statement:

The Hospital Management System (HMS) aims to streamline and automate hospital operations through a centralized, digital platform that integrates patient management, doctor scheduling, billing, and recordkeeping. By reducing manual paperwork and minimizing errors, HMS enhances efficiency and accuracy in healthcare administration. The system provides quick, secure access to critical data for authorized users, supporting improved decision-making, operational transparency, and better patient experiences. HMS targets hospitals seeking to modernize their processes while ensuring scalability and reliability for daily operations.

2. Stakeholder Identification

Category	Stakeholders	Interests / Concerns
Operational Stakeholders	Doctors, Nurses, Receptionists, Pharmacists	Efficient patient data access, appointment scheduling, reduced paperwork, accurate billing, and quick updates on records.
Executive Stakeholders	Hospital Administrator, IT Manager, Finance Manager	System performance, cost efficiency, data accuracy, compliance, and reporting capabilities.
External Stakeholders	Patients, Vendors (equipment/pharmacy suppliers)	Secure access to personal medical data, accurate billing, appointment booking, and reliable service.
Technical Stakeholders	Software Developers, System Administrators, Database Engineers	Reliable performance, maintainability, scalability, and data integrity of the HMS.

3. Functional Requirements

1. **Patient Management** – Register, edit, and delete patient records; view medical history.
2. **Doctor & Staff Management** – Maintain staff profiles, manage schedules, and assign duties.
3. **Appointment Scheduling** – Create, update, and cancel appointments; view doctor availability.
4. **Billing Module** – Generate, track, and print invoices for patient services.
5. **Report Generation** – Create summary reports on patients, doctors, and billing activities.
6. **Authentication and Authorization** – Role-based login for admins, doctors, and receptionists.

7. **Database Management** – Centralized storage of all hospital data with backup functionality.
8. **Pharmacy Management (Optional)** – Track prescribed medications and manage inventory levels.

4. Requirements Elicitation Techniques

Technique	Purpose	Justification
Interviews	Gather detailed, context-specific insights from hospital staff (doctors, receptionists, administrators).	Provides qualitative understanding of operational needs and pain points.
Questionnaires	Collect structured data from a large group of users (nurses, clerical staff, etc.).	Efficiently captures common requirements and usability expectations.
Observation	Observe daily workflows and bottlenecks in patient registration and appointment handling.	Ensures accurate understanding of current processes.
Document Analysis	Review existing paper forms and reports used in the hospital.	Helps design digital equivalents and ensures all required data fields are captured.

5. Interview Agendas

A. Internal Operational Stakeholder – Receptionist

Topics & Questions:

1. Describe your daily workflow for patient registration and appointment scheduling.
2. What challenges do you face with the current manual system?
3. How do you manage cancellations or rescheduling?
4. What features would make your work faster or more accurate?
5. How often do you interact with doctors or billing staff when handling patient data?
6. Do you face data duplication or loss?
7. How do you prefer the interface to be designed for ease of use?

B. Internal Executive Stakeholder – Hospital Administrator

Topics & Questions:

1. What are your primary goals for implementing an HMS?
2. What key metrics or reports are most important for decision-making?
3. How do you currently track staff performance and patient throughput?
4. What concerns do you have regarding data security and system maintenance?
5. What level of control do you need over system configurations and user permissions?
6. How would you measure the system's success after deployment?
7. Are there any compliance or audit requirements the system must satisfy?

6. Questionnaire for Operational Users

Please rate each statement on a scale of **1 (Strongly Disagree)** to **5 (Strongly Agree)**.

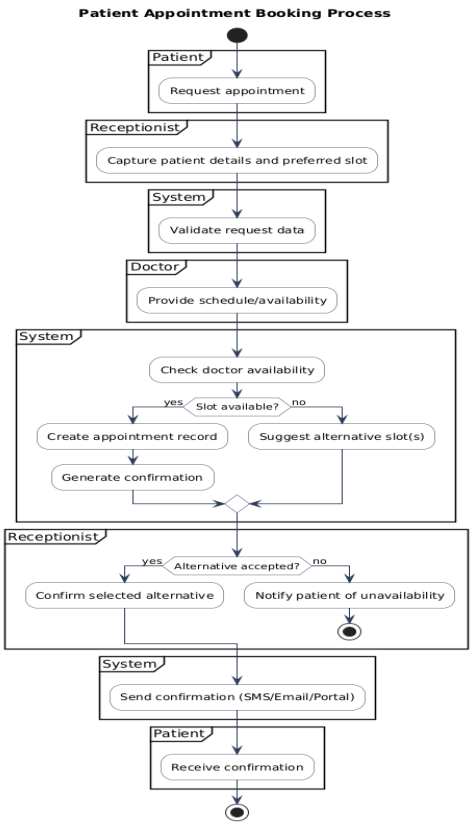
Question	Rating (1–5)
The current hospital management process is time-consuming.	
I frequently face issues retrieving patient information.	
I need a system that simplifies appointment scheduling.	
It's difficult to generate patient or billing reports manually.	
I would like automated alerts or reminders for appointments.	
The system should allow easy access to patient history.	
I prefer a user-friendly interface with minimal training required.	

Question	Rating (1–5)
Data security and privacy are my top concerns.	
The system should reduce paperwork and manual entries.	
I would recommend adopting a digital HMS in our hospital.	

7. Workflow Documentation (to be modeled in StarUML)

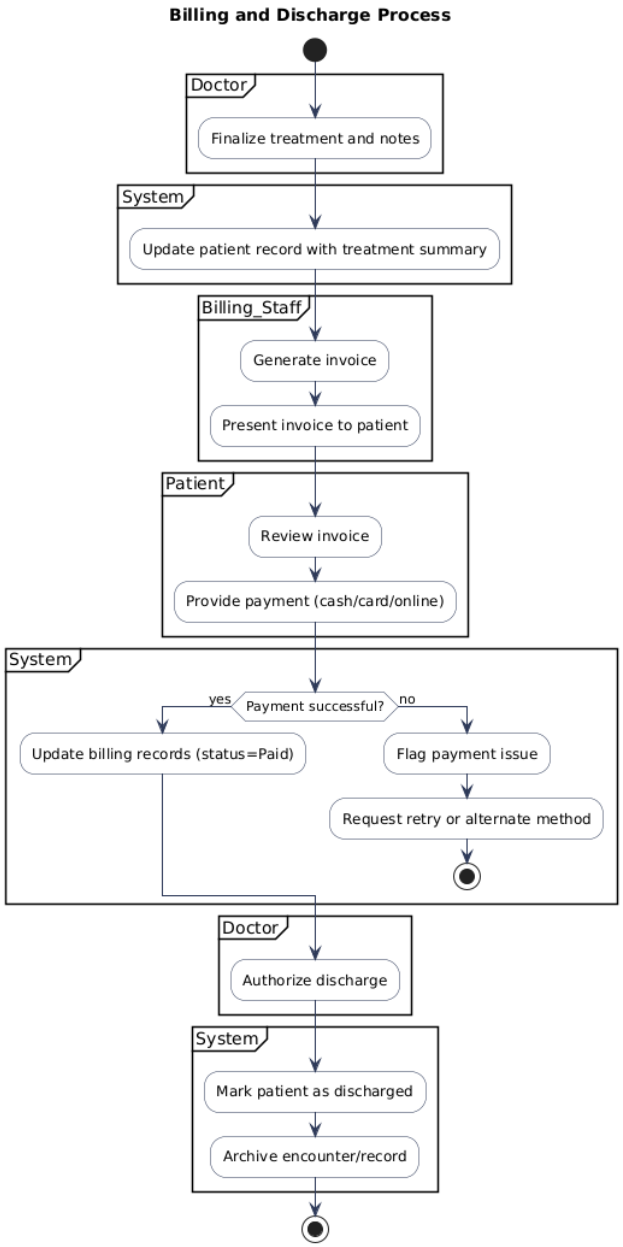
Activity Diagram 1: Patient Appointment Booking Process

- Steps:
  1. Patient requests appointment →
  2. Receptionist checks doctor's schedule →
  3. Appointment availability confirmed →
  4. Appointment recorded in HMS →
  5. Confirmation sent to patient.

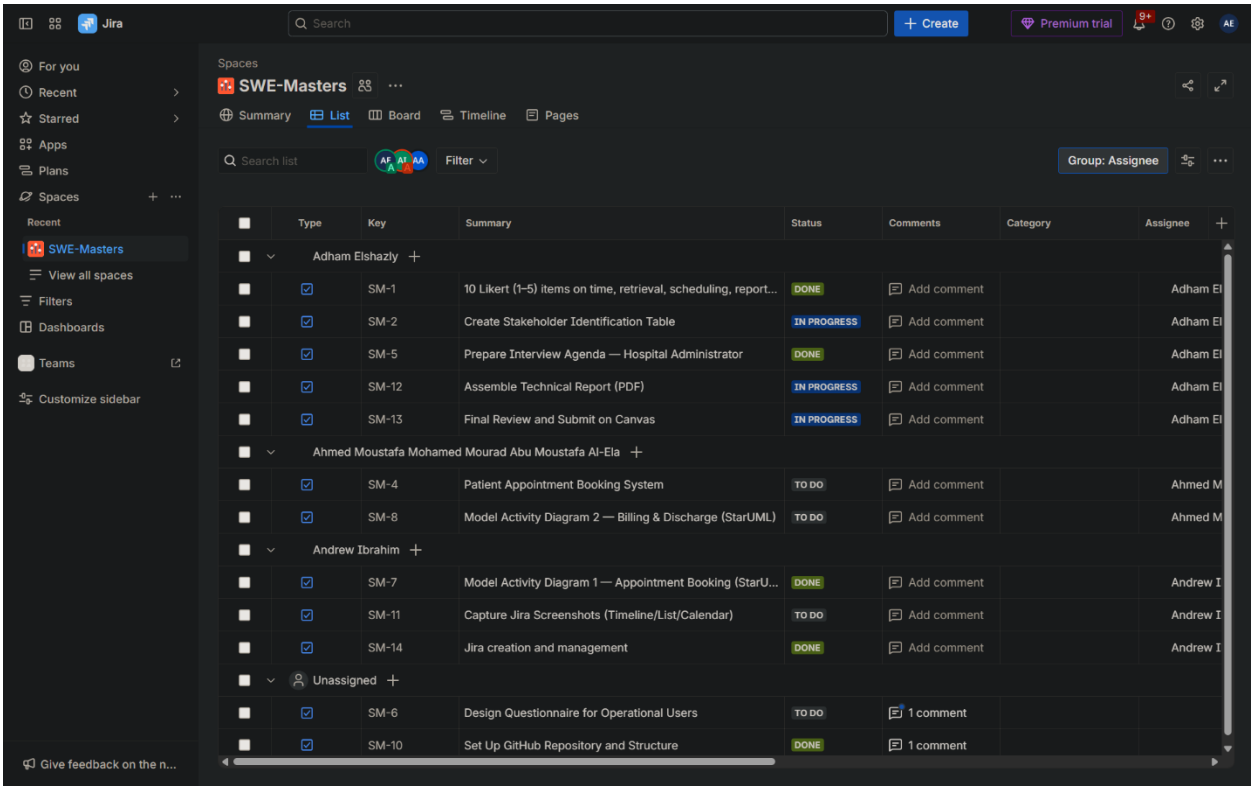


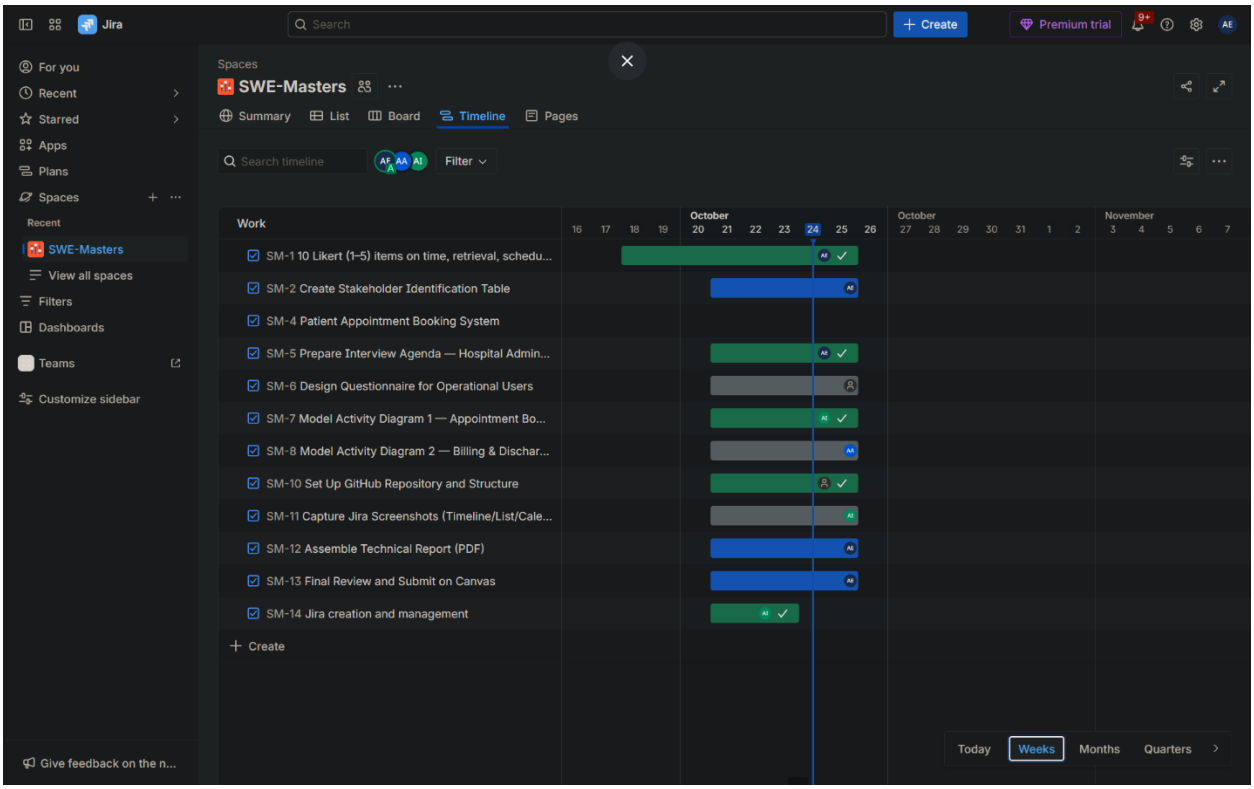
Activity Diagram 2: Billing and Discharge Process

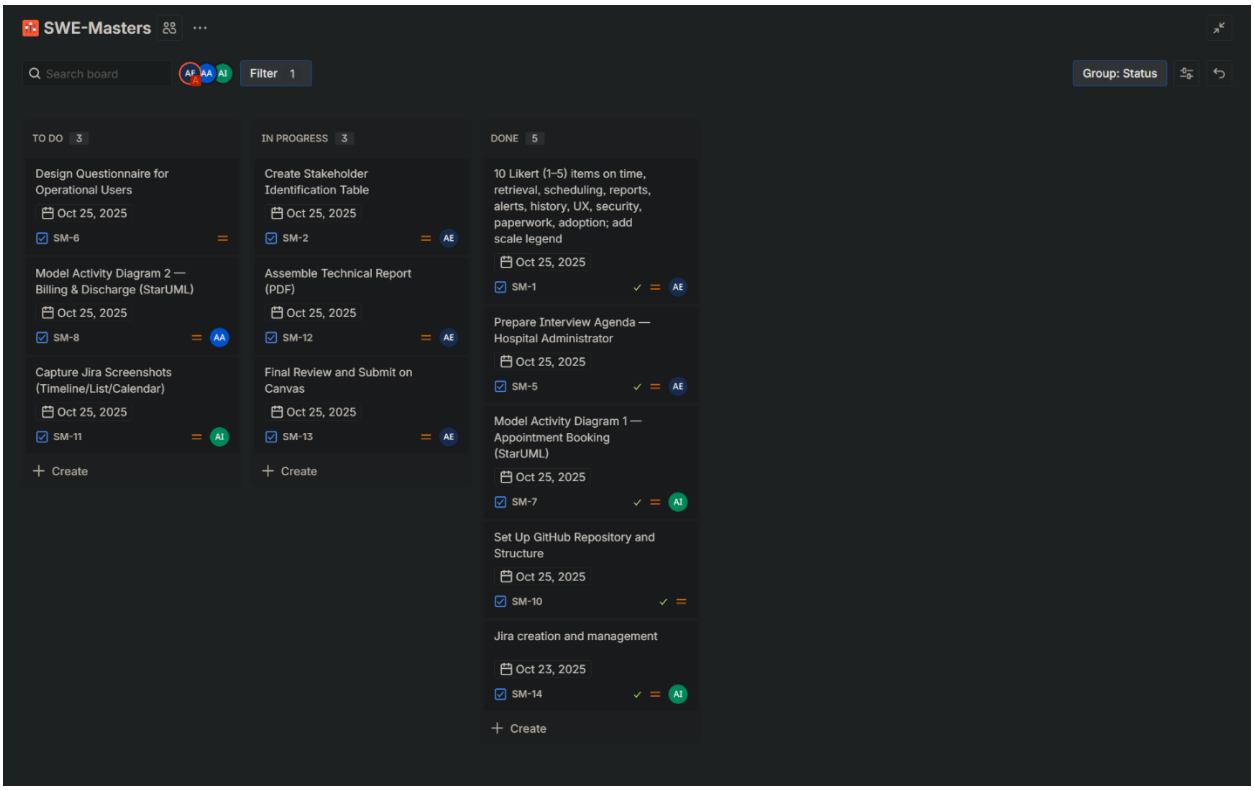
- Steps:
    1. Doctor finalizes treatment →
    2. Billing staff generates invoice →
    3. Patient reviews and pays bill →
    4. System updates billing records →
    5. Patient discharged and record archived.

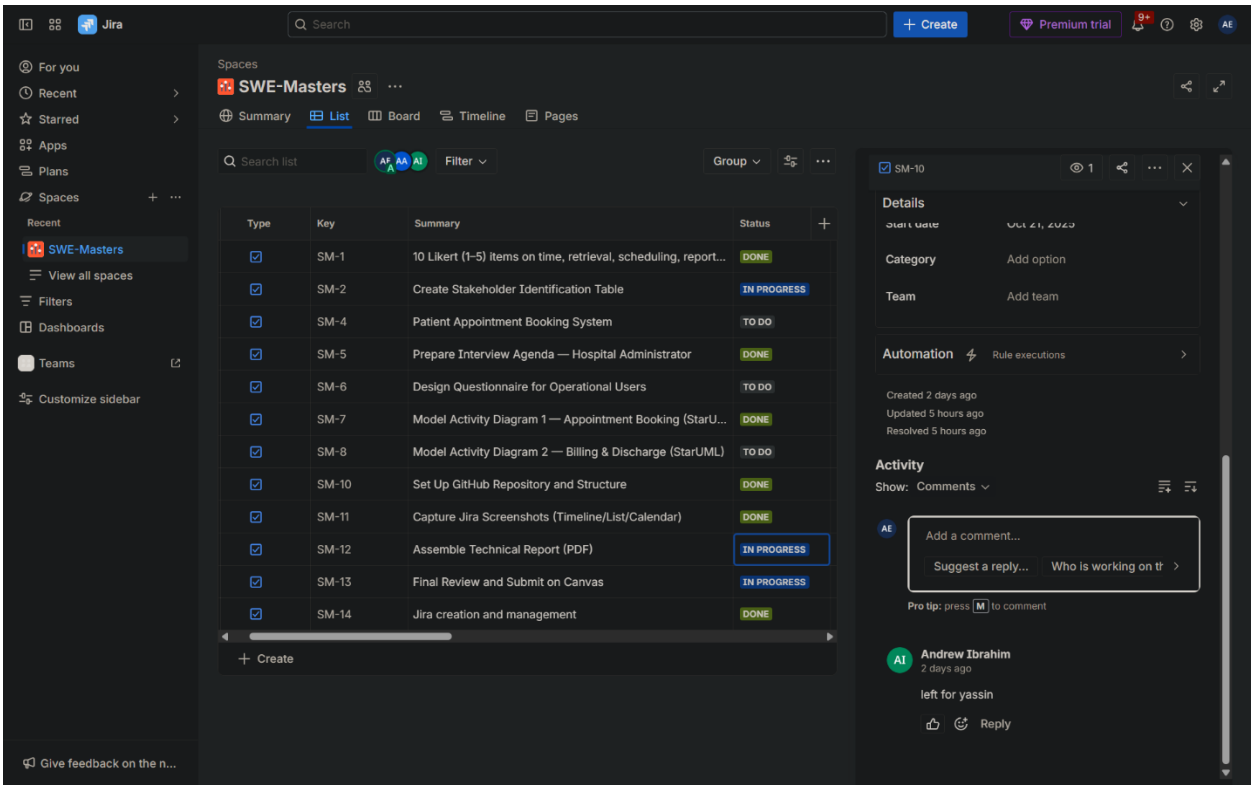


Jira Project Management dashboard




Jira
Search
Create
Premium trial
For you
Recent
Starred
Apps
Plans
Spaces
Recent
SWE-Masters
View all spaces
Filters
Dashboards
Teams
Customize sidebar
Give feedback on the n...
Spaces
SWE-Masters
Summary
List
Board
Timeline
Pages
Search timeline
Filter
Work
October
November
SM-1 10 Likert (1–5) items on time, retrieval, schedu...
SM-2 Create Stakeholder Identification Table
SM-4 Patient Appointment Booking System
SM-5 Prepare Interview Agenda — Hospital Admin...
SM-6 Design Questionnaire for Operational Users
SM-7 Model Activity Diagram 1 — Appointment Bo...
SM-8 Model Activity Diagram 2 — Billing & Dischar...
SM-10 Set Up GitHub Repository and Structure
SM-11 Capture Jira Screenshots (Timeline/List/Cale...
SM-12 Assemble Technical Report (PDF)
SM-13 Final Review and Submit on Canvas
SM-14 Jira creation and management
Create
Today
Weeks
Months
Quarters


SWE-Masters
Search board
Filter 1
Group: Status
TO DO 3
Design Questionnaire for Operational Users
Oct 25, 2025
SM-6
Model Activity Diagram 2 — Billing & Discharge (StarUML)
Oct 25, 2025
SM-8
Capture Jira Screenshots (Timeline/List/Calendar)
Oct 25, 2025
SM-11
Create
IN PROGRESS 3
Create Stakeholder Identification Table
Oct 25, 2025
SM-2
Assemble Technical Report (PDF)
Oct 25, 2025
SM-12
Final Review and Submit on Canvas
Oct 25, 2025
SM-13
Create
DONE 5
10 Likert (1–5) items on time, retrieval, scheduling, reports, alerts, history, UX, security, paperwork, adoption; add scale legend
Oct 25, 2025
SM-1
Prepare Interview Agenda — Hospital Administrator
Oct 25, 2025
SM-5
Model Activity Diagram 1 — Appointment Booking (StarUML)
Oct 25, 2025
SM-7
Set Up GitHub Repository and Structure
Oct 25, 2025
SM-10
Jira creation and management
Oct 23, 2025
SM-14
Create



Our team used Jira's Project Management template to organize and track all deliverable tasks. We created 15 tasks covering report sections, StarUML diagrams, and tool setup, assigning each to specific team members with clear start and due dates aligned to the October 25 deadline. We monitored progress daily using the List and Timeline views, updating task statuses from To Do to In Progress and Done. The Timeline view helped us visualize dependencies and ensure timely completion, while the List view provided a quick status overview. This approach ensured accountability, transparency, and on-time delivery of all components.

Assumptions

The following assumptions were made during the development of the **Hospital Management System (HMS)** vision and requirements:

1. **Notifications:**
   The system sends appointment confirmations via **SMS**, **Email**, or the **Patient Portal**.
2. **Pharmacy Module:**
   Pharmacy management is optional and considered **out of scope for Deliverable 1**.
3. **Role-Based Access:**
   The system includes distinct access permissions for the following roles: **Admin**, **Doctor**, **Receptionist**, **Nurse**, **Billing Staff**, and **Pharmacist**.
4. **Data Backup:**
   Automated daily backups are assumed to be configured and maintained by **system administrators**.
5. **Payment Methods:**
   The system supports **cash**, **card**, and **online payment** processing.
6. **Compliance:**
   The system must comply with applicable **healthcare data protection regulations** (e.g., **HIPAA** or local equivalents).